

SURIN AGROMET, Thailand

WMO: 484330

Lat: 14.891N

Lon: 103.452E

Elev: 145

StdP: 99.60

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.6	16.1	10.4	8.0	17.6	11.7	8.7	18.7	5.3	22.8	4.4	23.8	0.7	N/A	0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.6	38.2	26.3	36.9	26.0	35.8	25.9	27.9	33.7	27.6	33.2	27.2	32.7	1.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.6	22.5	30.7	26.2	22.0	30.3	25.8	21.5	29.9	90.4	33.9	88.6	33.3	87.1	32.8	34.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
(a)	(b)	(c)		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
4.6	3.6	3.2	DB	11.7	39.2	1.4	1.3	10.7	40.1	9.9	40.9	9.1	41.7	8.1	42.7	
			WB	10.5	29.7	1.5	2.1	9.4	31.2	8.5	32.3	7.6	33.5	6.5	35.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	24.2	26.4	29.0	30.4	30.0	29.6	28.9	28.6	28.2	27.6	26.5	24.3
	DBStd	2.77	2.64	2.81	2.50	2.09	1.47	1.14	1.26	0.99	1.21	1.13	1.88	2.60
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6494	440	460	588	612	620	587	584	576	545	546	494	442
	CDD18.3	3453	182	227	330	362	362	337	326	318	295	287	244	184
	CDH23.3	37953	1598	2419	4031	4641	4412	4003	3693	3471	3087	2802	2228	1567
	CDH26.7	16951	614	1171	2144	2549	2164	1808	1535	1352	1137	1018	868	591
(14) Wind	WSAvg	0.7	0.8	0.7	0.7	0.6	0.6	0.7	0.7	0.7	0.5	0.7	0.9	1.1
(15) Precipitation	PrecAvg	1289	2	10	23	83	173	166	185	209	284	127	25	3
	PrecMax	1626	16	63	52	226	451	295	343	369	519	319	81	27
	PrecMin	985	0	0	0	6	45	92	84	66	133	36	0	0
	PrecStd	180	4	16	16	61	106	62	68	85	119	79	26	7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	36.7	39.3	40.7	38.7	36.7	36.1	34.4	34.5	34.0	34.2	34.1
		MCWB	23.8	25.0	25.7	26.8	27.1	26.9	26.7	26.7	27.6	26.3	25.9	24.2
	2%	DB	32.7	35.5	37.6	38.8	36.9	35.3	34.6	33.5	33.2	32.8	33.0	32.7
		MCWB	23.4	24.4	25.3	26.6	26.8	26.9	26.8	26.6	26.9	25.9	25.3	24.1
	5%	DB	31.6	34.4	36.3	37.2	35.6	34.2	33.5	32.7	32.5	32.1	32.1	31.6
		MCWB	23.2	24.3	25.2	26.3	26.7	26.8	26.5	26.5	26.7	25.7	24.8	23.7
	10%	DB	30.2	32.8	35.0	35.8	34.4	33.2	32.4	31.9	31.6	31.3	31.2	30.2
		MCWB	22.5	23.8	25.1	26.0	26.6	26.6	26.3	26.4	26.5	25.6	24.5	23.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	26.5	26.9	28.2	28.1	28.3	28.2	27.7	28.2	27.4	26.8	25.5
		MCDB	31.5	33.9	35.4	36.2	34.9	34.2	33.4	32.2	32.7	31.9	32.3	31.9
	2%	WB	24.9	25.8	26.2	27.2	27.6	27.7	27.5	27.4	27.6	26.7	26.2	24.7
		MCDB	30.6	33.0	34.4	35.4	34.3	33.3	32.6	31.8	32.0	31.1	31.4	31.0
	5%	WB	24.1	25.2	25.9	26.8	27.2	27.2	27.0	27.0	27.1	26.4	25.7	24.2
		MCDB	29.3	32.1	34.0	34.6	33.5	32.6	31.9	31.4	31.4	30.5	30.5	30.0
	10%	WB	23.4	24.6	25.5	26.4	26.9	26.9	26.6	26.7	26.7	26.0	25.2	23.5
		MCDB	28.3	31.0	33.3	34.0	33.0	32.1	31.2	30.8	30.6	30.0	29.5	28.9
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.6	11.6	10.6	8.9	8.0	7.4	7.1	6.8	7.6	9.5	11.2
		MCDBR	12.7	13.6	13.0	12.4	10.7	9.2	8.9	8.0	8.1	8.7	10.2	11.8
		MCWBR	4.7	4.1	3.4	3.2	3.0	2.7	2.9	2.8	2.9	2.9	3.5	4.4
	5% WB	MCDBR	11.0	11.8	11.5	11.1	9.3	8.4	8.0	7.5	7.6	7.8	9.1	11.0
		MCWBR	4.5	3.8	3.5	3.4	3.1	3.0	3.1	3.0	3.0	2.9	3.4	4.4
(40) Clear-Sky Solar Irradiance	taub		0.498	0.613	0.663	0.600	0.500	0.467	0.460	0.463	0.497	0.523	0.475	0.466
	taud		2.032	1.758	1.689	1.872	2.189	2.291	2.309	2.291	2.185	2.081	2.194	2.163
	Ebn at Noon		793	719	697	743	808	826	834	841	816	783	810	812
	Edh at Noon		166	228	250	208	149	133	131	135	150	163	140	143
(44) All-Sky Solar Radiation	RadAvg		5.01	5.25	5.37	5.75	5.68	5.36	5.08	4.90	4.68	4.94	4.95	4.84
	RadStd		0.27	0.22	0.20	0.23	0.30	0.20	0.32	0.30	0.31	0.33	0.23	0.15

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	+0.26	N/A	N/A	N/A	N/A	+0.25	+0.22	N/A	N/A	+103	+103

Nomenclature: See separate page